Author Contributions Checklist Form

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Met opmerkingen [A1]: This document will be read by readers after an accepted manuscript is published so please phrase your responses for future readers and make sure to save your completed form as a PDF without these comments. Please see https://jasa-acs.github.io/repro-guide/pages/acc.html for details, links to examples, and information on how to save your completed form as a PDF. In particular make sure to remove these comments when saving as a PDF.

Part 1: Data

☐ This paper **does not** involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).

Met opmerkingen [A2]: If this box is checked and if no simulated/synthetic data files are provided by the authors, please skip directly to the Code section. Otherwise, continue.

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

Met opmerkingen [A3]: A short (< 100 words) description of the data. Additional details can be provided in README file(s) accompanying the materials.

Furthermore, we analyze an electrophysiological experiment that originates from session \texttt{715093703} of the Neuropixels extracellular recording collection maintained by the Allen Institute for Brain Science. Those data are publicly available.

Availability

☒ Data **are** publicly available

☐ Data **cannot be made** publicly available

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

☒ Data are available online at: https://brain-map.org/our-research/circuits-behavior

☐ Data are available as part of the paper's supplementary material.

☐ Data are publicly available by request, following the process described here:

Met opmerkingen [CP4]: If the data are available by request to the authors or some other data owner, please make sure to explain the process of requesting access to the data and the details of how to access the specific data files needed. Details are generally best provided in README file(s) accompanying the materials.

☐ Data are available through some other mechanism, described here:

Non-publicly available data

Discussion of lack of publicly available data:

Description

File format(s)

☐ CSV or other plain text:
☒ Software-specific binary format (.Rda, Python pickle, etc.): AllenSDK in Python
☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
☐ Other (described here):

Data dictionary

☐ Provided by the authors in the following file(s):
☐ Data file(s) is (are) self-describiing (e.g., netCDF, HDF5, Zarr files)
☒ Available at the following URL:

`https://allensdk.readthedocs.io/en/stable/visual_coding_neuropixels.html`

Additional information (optional)

Met opmerkingen [A5]: If the data for this manuscript are publicly available, skip to the Description section below. Otherwise, continue.

The Journal of the American Statistical Association requires authors to make data accompanying their papers available to the scientific community except in cases where: 1) public sharing of data would be impossible, 2) a synthetic/facsimile dataset in the same format as the real data is provided (recognizing that results may differ substantially from the real data analyses), and 3) the scientific value of the results and methods outweigh the lack of reproducibility.

Met opmerkingen [A6]: For example:
- why data sharing is not possible,
- what synthetic/facsimile data are provided, and
- why the value of the paper's scientific contribution outweighs the lack of reproducibility.

Met opmerkingen [CP7]: Check all that apply.

Met opmerkingen [CP8]: A data dictionary provides information that allows users to understand the meaning, format, and use of the data.

Met opmerkingen [CP9]: Provide any details that would be helpful in understanding the data. If relevant, provide unique identifier/DOI/version information and/or license/terms of use.

Part 2: Code

Abstract

The synthetic data are generated from spatiotemporal graphon Hawkes processes on ([0,1]), using eight Legendre polynomial basis functions for the spatial structure and three exponential basis functions for the temporal kernel. We consider four graphons within the specified model class and three perturbed variants to assess estimation accuracy, support recovery, and robustness to misspecification. Each experiment is replicated over 100 simulated sample paths, with varying observation horizons for the consistency analysis. The simulation and estimation procedures are implemented in Matlab, using seed 74973836 for reproducibility.

Met opmerkingen [A10]: A short (< 100 words) description of the code. Additional details can be provided in README file(s) accompany the materials. If no code is provided, please state this and say why (e.g., if the paper contains no computational work).

Description

Code format(s)

Met opmerkingen [CP11]: Check all that apply.

☒ Script files
- ☐ R ☐ Python ☐ Julia ☒ MATLAB
- ☐ Other:

☐ Package
- ☐ R ☐ Python ☐ Julia ☐ MATLAB toolbox
- ☐ Other:

☐ Reproducible report
- ☐ Quarto or R Markdown ☐ Jupyter notebook
- ☐ Other:

☐ Shell script

☐ Other (described here):

Supporting software requirements

Met opmerkingen [A12]: Please cite all software packages in the References Section in similar fashion to paper citations, citing packages that are foundational to the research outcome (including packages that implement methods to which you compare your methods). You may elect to not cite packages used for supporting purposes. For R packages, note that running `citation('name_of_package')` often shows how the package authors wish to be cited.

Version of primary software used

R2023a

Met opmerkingen [CP13]: List key software used here. Include version number(s) if not otherwise documentd in the material provided.

For example: R version 4.5.1.

Libraries and dependencies used by the code

Met opmerkingen [CP14]: List packages (e.g., R or Python packages) and libraries here, including version numbers (e.g., version numbers for any R or Python packages used) if not otherwise documented with the materials provided (e.g., an `renv` lock file or Conda environment file or in a README).

If documented with the materials, indicate here where to find the package and version information.

Supporting system/hardware requirements (optional)

Met opmerkingen [A15]: System/hardware requirements including operating system with version number, access to cluster, GPUs, etc.

Parallelization used

Met opmerkingen [A16]: For more complicated situations, more details can be provided in README file(s) accompanying the materials.

☐ No parallel code used
☒ Multi-core parallelization on a single machine/node
Number of cores used: 8
☐ Multi-machine/multi-node parallelization
Number of nodes and cores used:

License

☒ MIT License (default)
☐ BSD
☐ GPL v3.0
☐ Creative Commons
☐ Other (described here):

Additional information (optional)

Met opmerkingen [CP17]: By default, supplementary material will be archived on figshare by JASA. Authors are strongly encouraged to also make their code available in a public code repository, such as on GitHub, GitLab, or BitBucket. If relevant, please provide unique identifier/DOI/version information (e.g., a Git commit ID, branch, release, or tag).

If the code and workflow are provided together, this section may be omitted, with information provided in the "Location" section below.

Part 3: Reproducibility workflow

Met opmerkingen [CP18]: The materials provided should provide a straightforward way for reviewers and readers to reproduce analyses with as few steps as possible.

Scope

The provided workflow reproduces:

☒ Any numbers provided in text in the paper

☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))

☒ All tables and figures in the paper

☐ Selected tables and figures in the paper, as explained and justified here:

Workflow details

Location

Met opmerkingen [CP19]: Check all that apply, and in the case of a Git repository include unique identifier, such as specific commit ID, branch, release, or tag.

Indicate where the materials (generally including the code, unless in a separate location and indicated in the previous section) are available.

We strongly encourage authors to place their materials (but not large datasets) in a Git repository hosted on a site such as GitHub, GitLab, or BitBucket. If the repository is private during the review process, please indicate the location where it will be available publicly upon publication.

If materials are provided in a Git repository, to ensure long-term availability, make sure that you have also either included the materials in the supplementary material (e.g., as a zip file obtained directly from the Git hosting service) or placed them in a permanent archive with a DOI, such as in Zenodo.

The workflow is available:

☐ As part of the paper's supplementary material

☒ In this Git repository: https://github.com/JRB-UvA/Graphon_estimation_reproducibility

☐ Other:

Format(s)

Met opmerkingen [CP20]: Check all that apply

☐ Single master code file

☐ Wrapper (shell) script(s)

☐ Self-contained Quarto, R Markdown file, Jupyter notebook, or other literate programming approach

☒ Text file (e.g., a readme-style file) that documents workflow

☐ Makefile

☐ Other (more detail in 'Instructions' below)

Instructions

To simulate the 100 sample paths of 7 different graphon Hawkes processes (the unperturbed and 3 perturbed processes referred to in the document), run parallel_estimation_legendre_complete.m
For each of those 7 sets of 100 realized paths, run lines 213—241 of the same code to reproduce the contents of Appendix C, as well as the numbers in Table 2 of the main text.
Table 1 can be reproduced by running parallel_estimation_legendre.m for the different values of T. Note that the user can alter commands like
%delete(gcp('nocreate'));
%parpool(7);
To prevent an overflow of RAM. If there is no issue with RAM overflow, you could simply surpress.

The data for the session 715093703 of the Neuropixels extracellular recording collection maintained by the Allen Institute for Brain Science can be found through the website of Allen Institute, yielding ecephys_session_715093703.nwb. We used Python to extract an .m file from this. The resulting .m file, consisting of the events in the time interval [5000;5250] is found in Allen_neuropixel.zip (unzip to get Allen_neuropixel.m).

Figure 1 can be reproduced by running allen_brain_legendre.m (for Legendre basis) and allen_brain.m (for Haar basis). Lines 271—288 and 172—189, respectively, recreate the required figures.
Finally, Figure A3 can be reproduced by running allen_brain.m with eta=0.

Met opmerkingen [A21]: Describe how to use the materials provided to reproduce analyses in the manuscript. For more complicated workflows, it is best to provide a short summary here with full details provided in README file(s) accompanying the reproducibility materials. If no workflow is provided, please state this and say why (e.g., if the paper contains no computational work).

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:
☐ <1 minute
☐ 1-10 minutes
☐ 10-60 minutes
☒ 1-8 hours
☐ >8 hours
☐ Not feasible to run on a desktop machine, as described here:

Met opmerkingen [A22]: For more complicated situations, including cases where run-time varies greatly across different analyses, more details can be provided in README file(s) that accompany the materials.

Additional documentation (optional)

Met opmerkingen [A23]: Additional documentation provided (e.g., R package vignettes, demos or other examples) that show how to use the provided code/software in other settings.

Notes (optional)

Met opmerkingen [A24]: Any other relevant information not covered on this form.